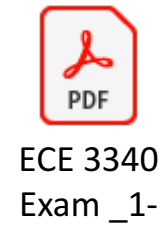


Exam 1 File

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1. (10 pts) Demonstrate the following 8-bit binary unsigned integer operation. Provide the corresponding operands and the results in decimal. Please show your work.

$1010\ 0101 \times 10 =$

binary	decimal
(1) 0100 1010 2*8+64=74	165*2 = 330 330%256 = 74

2. (10 pts) Convert the following 8-bit signed integers into decimal using Two's complement or convert the following decimal into 8-bit signed integers using Two's complement. Please show your work.

$1011\ 1000$
 $= -2^7 + 2^5 + 2^4 + 2^3 = -128 + 32 + 16 + 8 = -72$

 -85
 $85 = 64 + 16 + 4 + 1 = 2^6 + 2^4 + 2^2 + 2^0 = 01010101$
 $-85 = 10101010 + 1 = 10101011$

3. (8 pts) Convert the numbers into the specified number system.
1) Convert $(1023)_5$ to base-8 (Radix = 8);
 $1*5^3 + 2*5^1 + 3 = 138 \rightarrow (212)_8$

2) Convert $(762)_8$ to base-5;
 $7*8^2 + 6*8 + 2 = 498 \rightarrow (3443)_5$

4. (10 pts) Convert the following values (unsigned) to decimal. Please show your work.

$0111.1011 = 7.6875$

$10101.101 = 21.625$

5. (10 pts) Assume an 8-bit floating point format with a 4-bit exponent and a bias of 7 and implied 1. Provide the decimal representations of the following stored values.

Sign	Exponent	Mantissa
1	1 1 0 0	0 1 0 1

$-1.101*2^5 = -110100 = -52$

Sign	Exponent	Mantissa
0	0 1 0 1	0 1 0 0

$1.01*2^{-2} = 0.0101 = 0.3125$

6. (5 pts) What is the computational complexity of multiplying two $n \times n$ matrices using a naïve algorithm?
 $O(n^3)$

7. (12 pts) Reformulate the following expression to deal with catastrophic cancellation and test the result using 3 digits of precision for $x = 0.01$.

$\sqrt[4]{1+x} - \cos(x)$

Exact Result (using calculator or computer)
0.00254

Reformulation

$$\sqrt[4]{1+x} - \cos(x) = \frac{\sqrt[4]{1+x} - \cos^2 x}{\sqrt[4]{1+x} + \cos(x)} = \frac{\sqrt[4]{1+x} - 1}{\sqrt[4]{1+x} + \cos(x)} + \frac{\sin^2 x}{\sqrt[4]{1+x} + \cos(x)}$$
$$= \frac{x}{(\sqrt[4]{1+x} + \cos(x))(\sqrt[4]{1+x} + 1)} + \frac{\sin^2 x}{\sqrt[4]{1+x} + \cos(x)}$$

Original result	Reformulated result
$1.00 - 1.00 = 0$	1.00, 1.00, 1.00 $= x/4 + (\sin x)^2/2 =$ 0.0025+0.00005=0.00255

Relative error	Relative error
100%	0.394%

8. (10 pts) Reformulate using the Mean Value Theorem and test the result using 3 digits of precision for $x = 0.01$. Show your work.

$\sqrt{\pi+x} - \sqrt{\pi}$

Exact Result (using calculator or computer)
0.00282

Reformulation

$f(x) = x^{1/2}, f'(x) = 1/2 * x^{-1/2}$

$\rightarrow (\pi+x)^{1/2} - (\pi)^{1/2} = x^{1/2} * (\pi+x/2)^{-1/2} = x/2 * (\pi+x/2)^{-1/2}$

Original result	Reformulated result
$1.77 - 1.77 = 0$	$3.15^{1/2} = 1.77$ $1/1.77 = 0.565$ $0.005 * 0.565 = 0.00283$
Relative error 100%	Relative error 0.355%

9. (10 pts) Use a Taylor series expansion to reformulate the following numerical problem. Keep terms up to and including $O(x^3)$. Calculate the relative error before and after reformulation for $x = 0.02$. Use 3 significant digits for each operation.

$\frac{1 - \cos x}{x}$

Exact Result (using calculator or computer)
0.00100

Reformulation

$\cos(x) = \cos(0) - \sin(0)*x - \cos(0)/2*x^2 + \sin(0)/6*x^3$
 $= 1 - x^2/2$
 $\rightarrow x^2/2/x = x/2$

Original result	Reformulated result
0	0.00100
Relative error 100%	Relative error 0%

10. (15 pts) Use synthetic division to compute $f(x) = e^{-x} \sin x^2$ for $x = 0.1$ with an error bound of $O(x^4)$ using 3 significant digits. Show this calculation using synthetic division.

$e^x = 1 + x + 1/2 * x^2 + 1/6 * x^3$
 $e^{-x} = 1 - x + 1/2 * x^2 - 1/6 * x^3$
 $\sin x = \sin(0) + \cos(0)x$
 $\sin x^2 = x^2$

$f(x) = x^2 - x^3$

$$\begin{array}{r|rrrr} 0.1 & -1 & 1 & 0 & 0 \\ & & -0.1 & 0.09 & 0.00900 \\ \hline & -1 & 0.9 & 0.09 & 0.00900 \end{array}$$